

COURIER MANAGEMENT SYSTEM

A Salesforce-Based Shipment Booking, Tracking and Delivery
Management Solution

Project Documentation

User Story

Courier and logistics businesses find it hard to run shipments and deliveries smoothly because most of their work still happens on paper and in separate tools. With no single place to follow a parcel, staff cannot see where it is, delivery updates reach customers late, and customers are left without clear information.

Delivery agents are assigned manually, which leads to overlapping schedules and delays. Customers are not informed automatically, and managers have no performance figures to judge how well the operation is running. When several branches are involved, coordination becomes harder still. Together these problems cause unhappy customers, lower productivity, wasted effort and lost revenue.

1. Introduction

1.1 Project Overview

Many courier and logistics firms still depend on paper registers, spreadsheets and phone calls to run daily work. Parcel data sits in different places, nobody can see where a consignment is at a given moment, and customers keep calling to ask for updates. Assigning a delivery to an agent is done by hand, so double bookings and missed slots are common, and managers have no reliable numbers to judge how their branches are performing.

The Courier Management System (CMS) addresses these gaps with a cloud application built on the Salesforce Lightning platform. It keeps customers, shipments, delivery agents, branches, status updates and invoices in one connected data model. Flows and validation rules automate the routine steps, role-based security controls who can see what, and reports and dashboards turn raw records into information that managers can act on.

1.2 Purpose

The aim of this project is to replace scattered manual methods with a single, automated platform for the full courier cycle. It is meant to record parcel bookings in a standard, validated format, give staff a live view of each shipment's status, reduce manual effort in assigning agents and updating statuses, link invoices to the shipments they belong to, and provide branch-wise and agent-wise insight through analytics.

By bringing every branch onto one platform, CMS improves visibility, communication and speed of decisions, and it can grow with the business without a redesign. In the end, the goal is better service, happier customers and higher profitability.

2. Ideation Phase

2.1 Problem Statements

Two key stakeholders were considered: the branch manager, who runs daily operations, and the customer, who depends on timely information about a parcel.

ID	Stakeholder	Goal	Obstacle	Root Cause	Feeling
PS-1	Branch Manager	Handle shipments smoothly	Tracking is done by hand	Tools and records are not connected	Stressed
PS-2	Customer	Follow the parcel's progress	Status updates arrive late	No single system holds shipment data	Uncertain

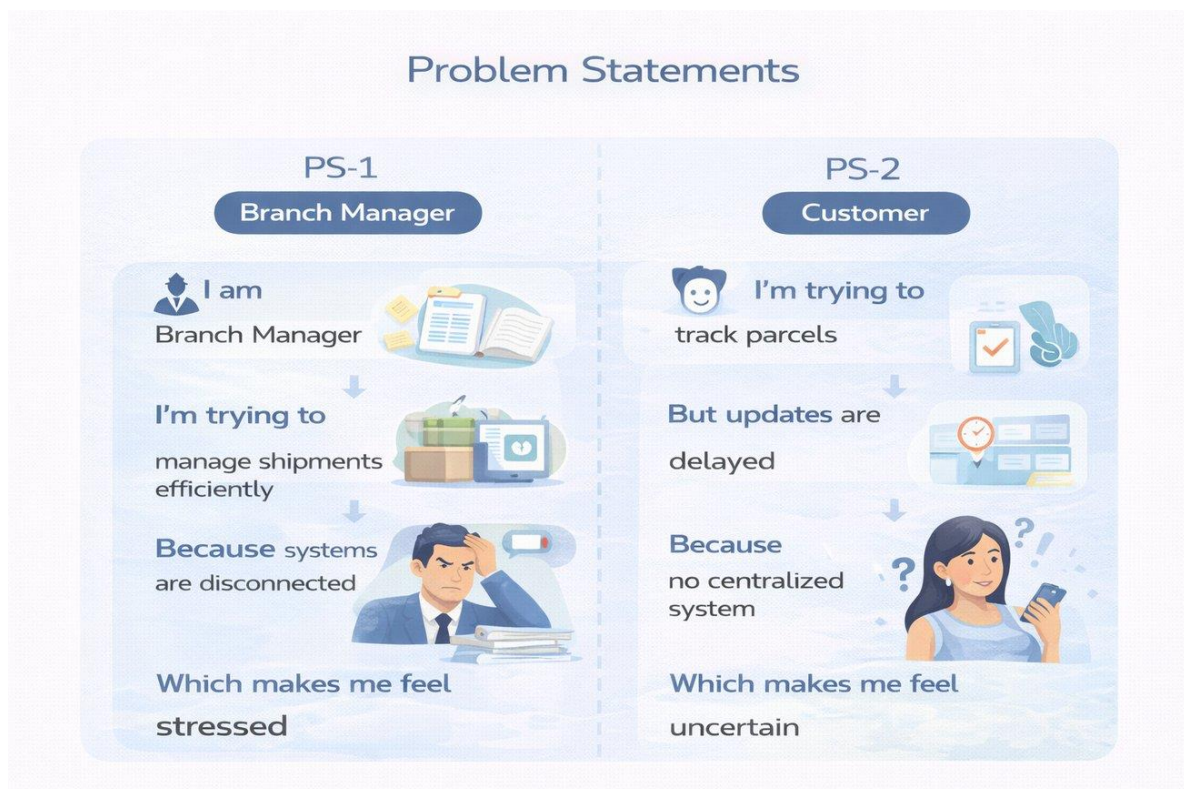


Figure 1: Problem statements for the branch manager and the customer

2.2 Empathy Map Canvas

User Persona: Branch Manager

- **Think & Feel:** Worried about late deliveries and data-entry mistakes
- **Hear:** Repeated complaints and follow-up calls from customers
- **See:** Paper registers and several unconnected spreadsheets
- **Say & Do:** Chases agents and updates records manually
- **Pain:** Clashing delivery schedules and updates that never get recorded

- **Gain:** Automatic tracking and a dashboard that shows the whole picture

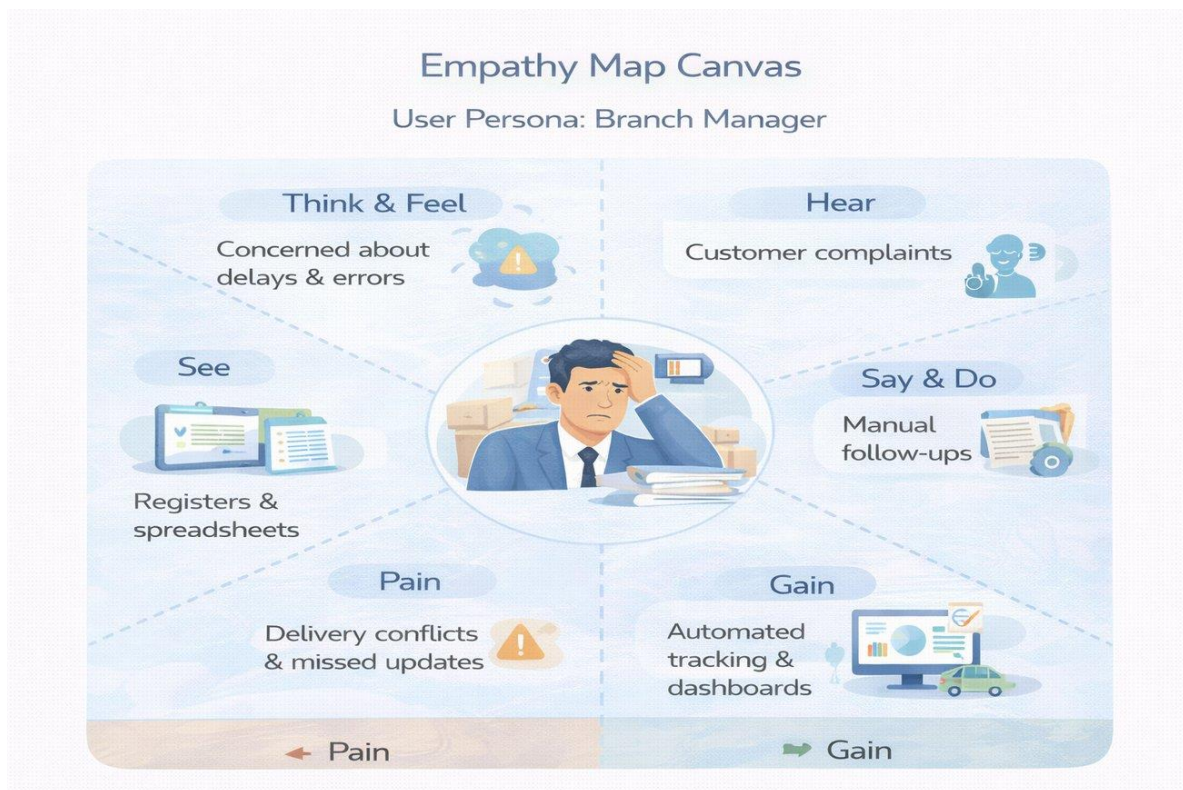


Figure 2: Empathy map for the branch manager persona

2.3 Brainstorming & Idea Prioritization

Ideas Considered

- Paper registers
- Spreadsheet tracking
- Standalone web portal
- Salesforce CRM

Final Selection

Salesforce CRM was selected because of its:

- Built-in automation
- Ability to scale as branches are added
- Real-time visibility through reports and dashboards

2.3 Brainstorming & Idea Prioritization – Courier Management System

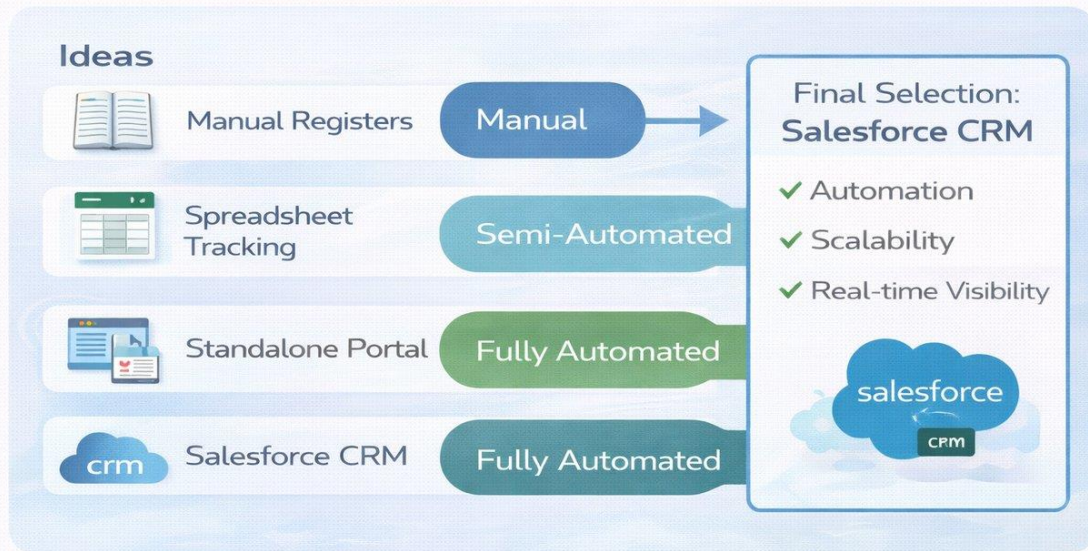


Figure 3: Options considered and the final selection

3. Requirement Analysis Phase

3.1 Customer Journey Map

A parcel moves through these stages: Booking → Shipment creation → Agent assignment → Transit → Delivery update → Invoice → Reporting.

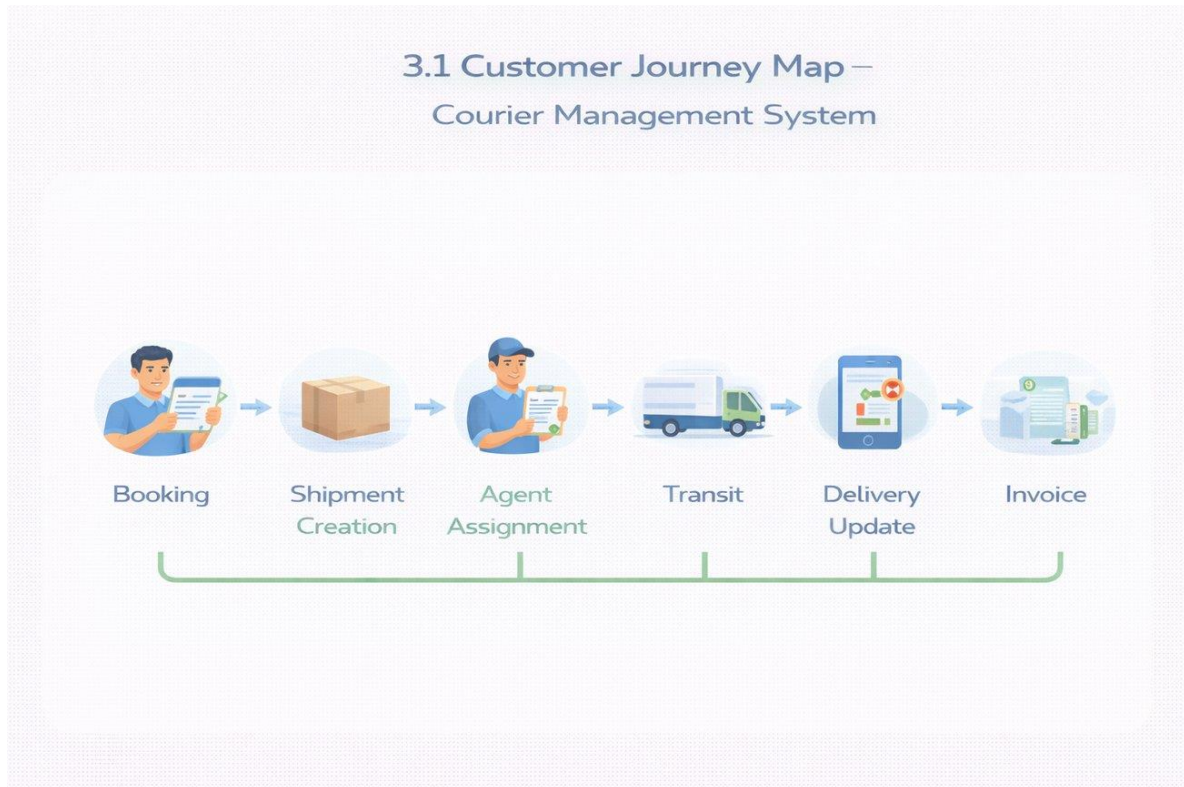


Figure 4: Customer journey through the courier process

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation
FR-6	Reports and dashboards

Non-Functional Requirements

NFR	Description
Usability	Clean Lightning interface that staff can learn quickly
Security	Role-based access control

NFR	Description
Reliability	Automation that runs consistently without manual correction
Performance	Status changes visible almost instantly
Availability	Cloud access at any time, from anywhere
Scalability	Supports a growing number of branches

3.3 Data Flow Diagram

Flow: Customer input → Shipment creation → Validation → Agent assignment → Delivery update → Invoice → Reports.

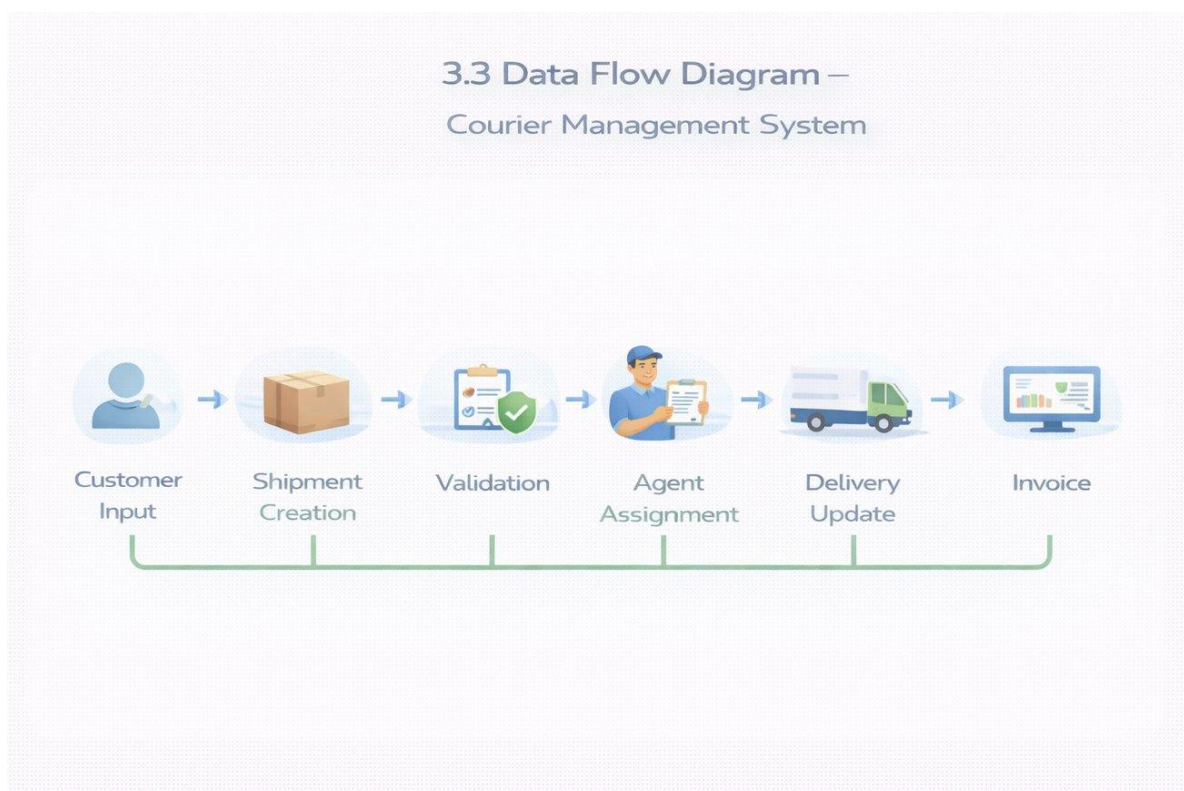


Figure 5: Data flow of the courier system

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports & Dashboards

4. Project Design Phase

4.1 Problem–Solution Fit

CMS removes the fragmentation in courier operations by automating shipment tracking, delivery assignment, billing and reporting on one central Salesforce platform. Each weakness of the manual process has a matching feature: registers are replaced by a central dashboard, delayed updates by automated tracking, scattered sheets by agent assignment, and confused customers by visible billing and status information.

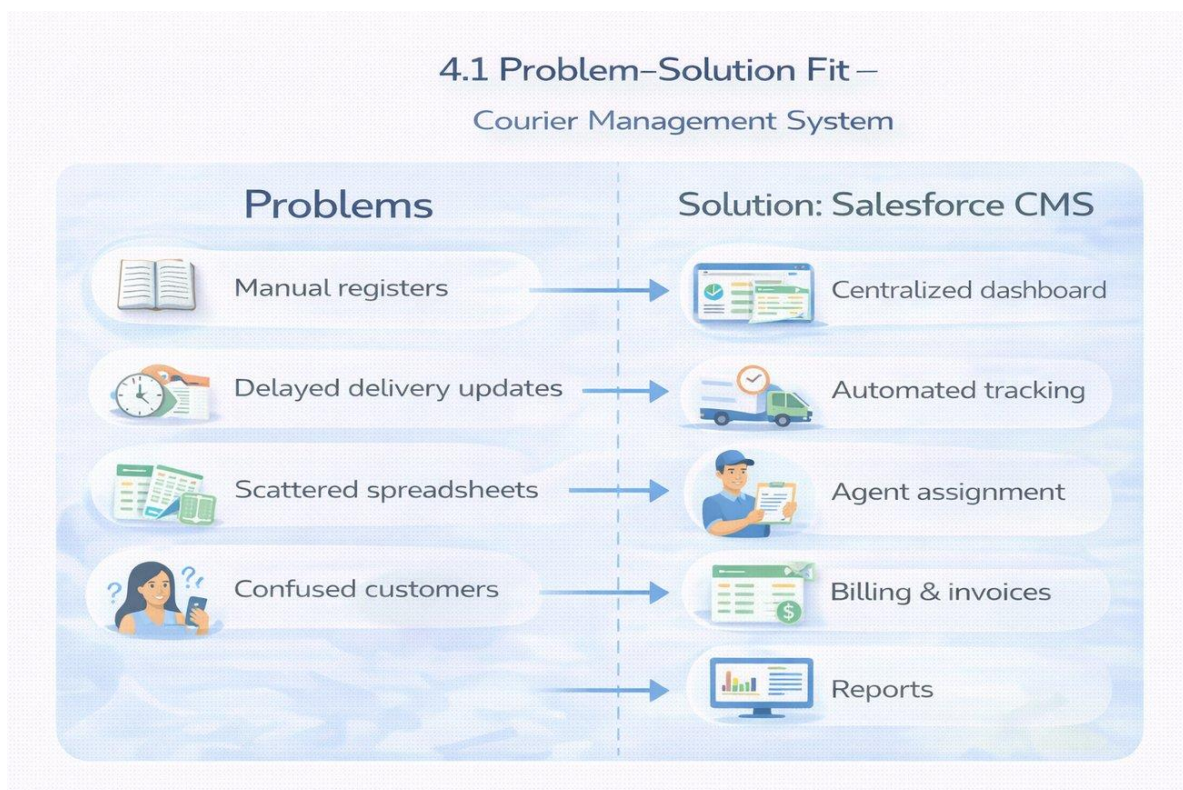


Figure 6: Mapping of current problems to CMS features

4.2 Proposed Solution

Parameter	Description
Problem	Manual shipment management
Solution	Salesforce-based Courier Management System
Innovation	Automated tracking and billing
Impact	Faster deliveries
Business Model	Courier services
Scalability	Ready for multiple branches

4.3 Solution Architecture

Components of the solution:

- Customer object
- Shipment object
- Delivery Agent object
- Branch object
- Delivery Update object
- Invoice object
- Flows
- Reports
- Dashboards

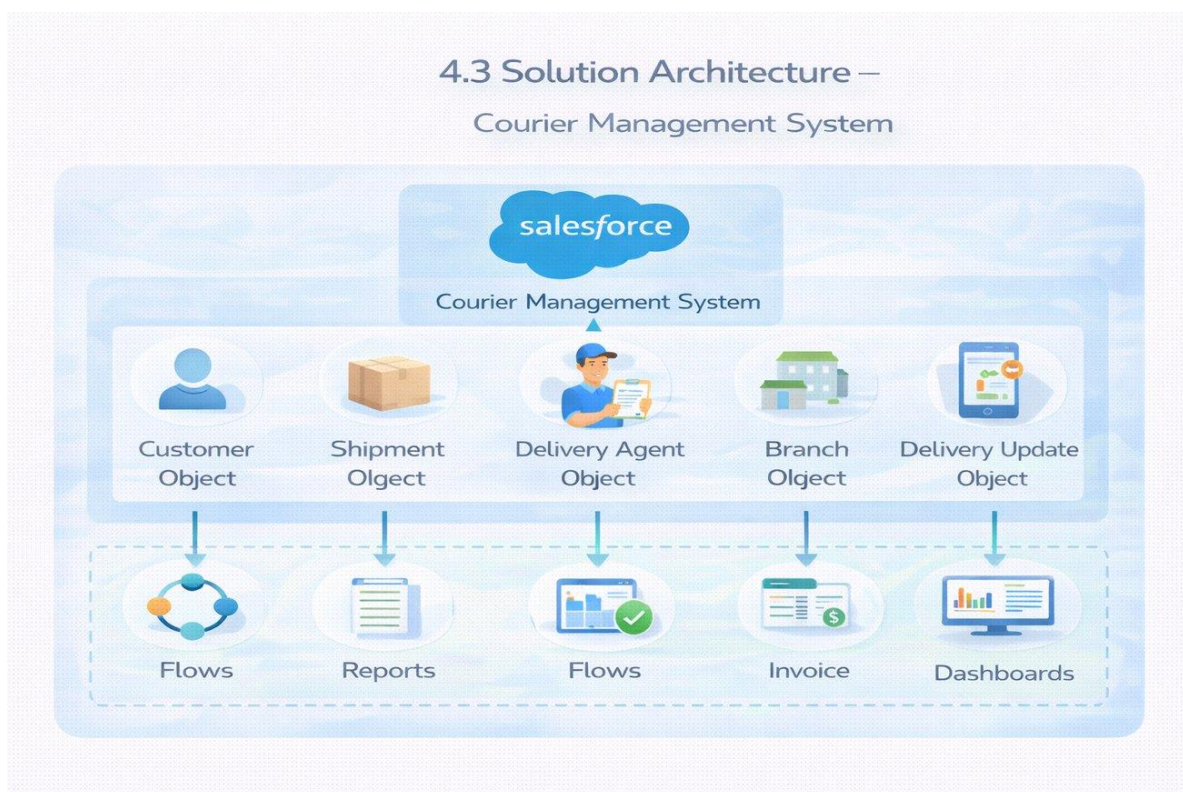


Figure 7: Solution architecture of CMS

5. Project Planning & Scheduling

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule and Estimation

Sprint	Epic	ID	User Story / Task	Pts	Priority	Member
1	Developer setup	USN-1	As an administrator, I want to create a Salesforce Developer org so that I can build and test the courier application.	3	High	Member 1
2	Data modeling	USN-2	As an admin, I want to create Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice objects with relationships so that data is stored in an organised way.	5	High	Member 2
2	Data modeling	USN-3	As a staff member, I want tabs and a Lightning app so that I can move between courier modules easily.	5	High	Member 3
3	Automation	USN-4	As the system, I want to check dates, weights, prices and mandatory fields so that bad data cannot be saved.	3	High	Member 4
3	Automation	USN-5	As a dispatcher, I want shipment status to update automatically from delivery updates so that work is faster and manual steps are reduced.	3	High	Member 4
4	Security	USN-6	As an administrator, I want profiles, roles and users so that shipment and billing data is available only to the right people.	5	High	Member 2

Sprint	Epic	ID	User Story / Task	Pts	Priority	Member
5	Reports	USN-7	As a branch manager, I want reports on shipment status, agent performance and revenue so that I can monitor operations.	4	High	Member 3
6	Dashboards	USN-8	As a branch manager, I want a dashboard that combines key reports so that I can review performance at a glance.	4	Medium	Member 4

Project Tracker, Velocity & Burndown

Dates below are indicative; adjust them to match your actual schedule.

Sprint	Total Points	Days	Start	Planned End	Points Done	Release Date
Sprint-1	3	6 days	05 Oct 2026	10 Oct 2026	3	10 Oct 2026
Sprint-2	10	6 days	12 Oct 2026	17 Oct 2026	10	17 Oct 2026
Sprint-3	6	6 days	19 Oct 2026	24 Oct 2026	6	24 Oct 2026
Sprint-4	5	6 days	26 Oct 2026	31 Oct 2026	5	31 Oct 2026
Sprint-5	4	6 days	02 Nov 2026	07 Nov 2026	4	07 Nov 2026
Sprint-6	4	6 days	09 Nov 2026	14 Nov 2026	4	14 Nov 2026

Total planned work is 32 story points over 6 sprints, giving an average velocity of about 5.3 points per sprint ($32 \div 6$).

6. Project Development Phase

What you will learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts
4. Writing validation rules
5. Building automation with flows
6. Developing reports and dashboards
7. Setting up security with profiles, roles and users
8. End-to-end application testing

Milestone 1: Salesforce Account

Introduction

If you are new to Salesforce and unsure where to begin, this project shows how a real business application, in this case a courier management system, can be built on the platform without writing code. Salesforce offers tools that automate processes, improve accuracy and manage customer operations with ease. Here they are used to streamline the courier workflow from parcel booking to delivery, billing and reporting.

What is Salesforce?

Salesforce is a customer success platform that helps organisations sell, serve, market, analyse and connect with their customers. It lets you manage relationships with customers and partners, collaborate with employees, and store data securely in the cloud, all accessible from anywhere.

Activity 1: Creating a Developer Account

A developer org is a free personal Salesforce environment used for learning and building.

1. Open <https://developer.salesforce.com/signup> in your browser.
2. On the sign-up form enter your first name and last name.
3. Enter your email address.
4. Set Job Title to Developer.
5. Set Company to your college name.
6. Set Country to India.
7. Accept the agreement and click Sign me up.

Build enterprise-quality apps fast to bring your ideas to life

- Build apps fast with drag and drop tools
- Customize your data model with clicks
- Go further with Apex code
- Integrate with anything using powerful APIs
- Stay protected with enterprise-grade security
- Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*
Your first name

Last Name*
Your last name

Email*
Your email address

Role*
Your job role

Company*
Company Name

Figure 8: Developer Edition sign-up form

Activity 2: Account Activation

1. Open the inbox of the email address used during sign-up. The message can take between 10 minutes and 2 hours to arrive.
2. Click Verify Account in the email.
3. Choose a password and answer the security question, then click Change Password.
4. You are redirected to the Salesforce Setup home page.

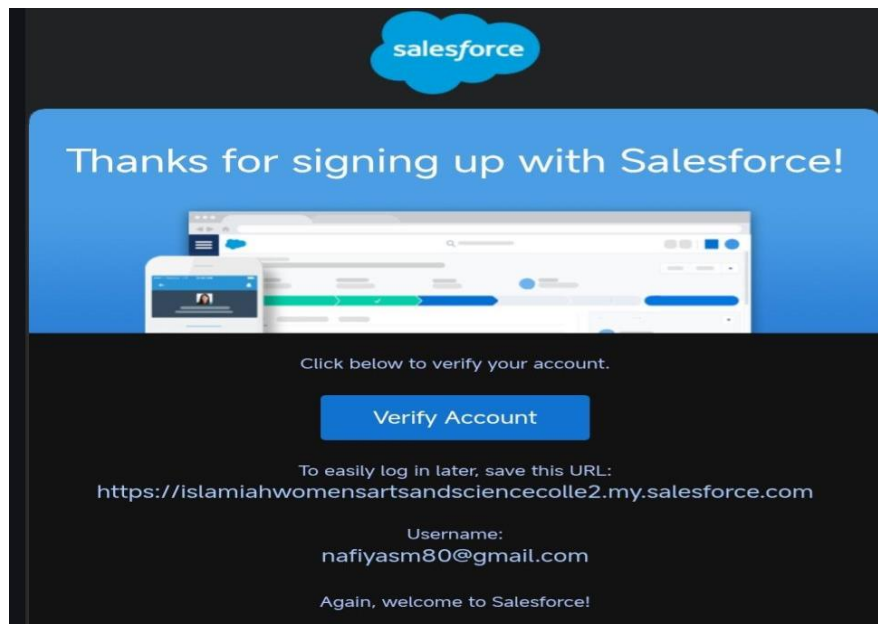


Figure 9: Verification email from Salesforce

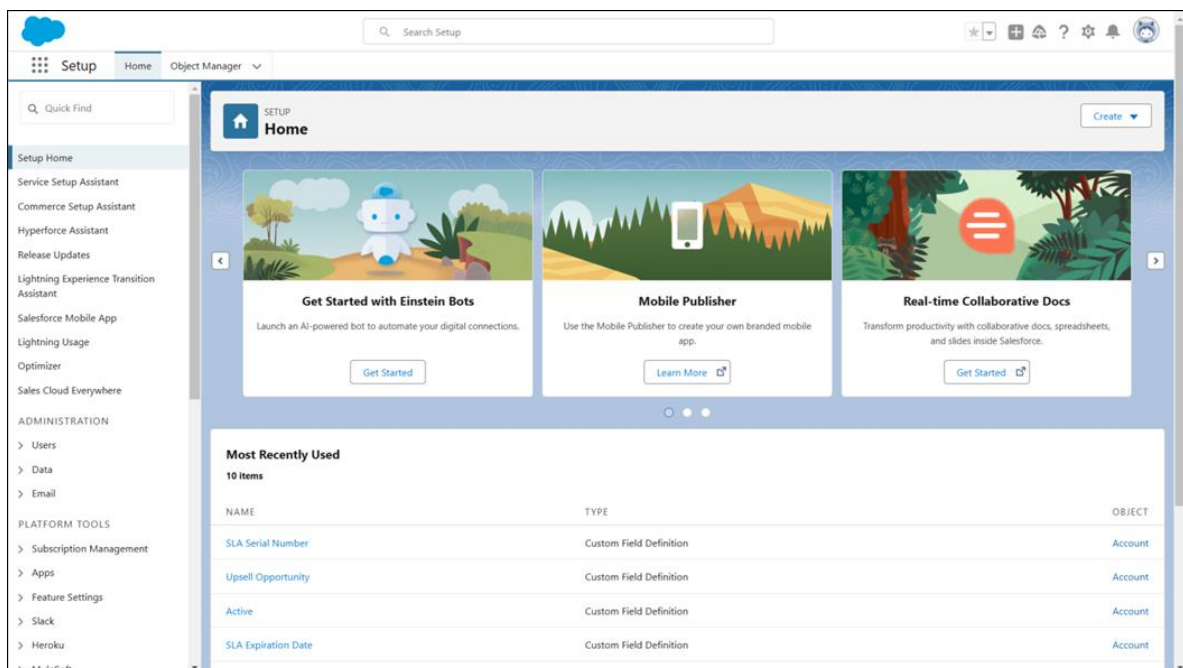


Figure 10: Salesforce Setup home page

Milestone 2: Objects

What is an object?

Objects in Salesforce are database tables that store data specific to an organisation.

An object is made up of fields (columns) and records (rows).

Custom objects created

1. **Customer__c** – stores personal and contact details of customers
2. **Shipment__c** – holds parcel details, source, destination and tracking information
3. **Delivery_Agent__c** – keeps delivery agent profiles and branch details
4. **Branch__c** – represents the courier branch offices
5. **Delivery_Update__c** – records each status update of a shipment
6. **Invoice__c** – manages billing details for delivered shipments

Relationships

Child Object	Parent Object	Type
Shipment	Customer	Lookup
Shipment	Delivery Agent	Lookup
Delivery Agent	Branch	Lookup
Delivery Update	Shipment	Master-Detail
Invoice	Shipment	Lookup

Steps to Implement

Activity 1: Create the Customer object

1. Click the gear icon → Setup → Object Manager → Create → Custom Object.

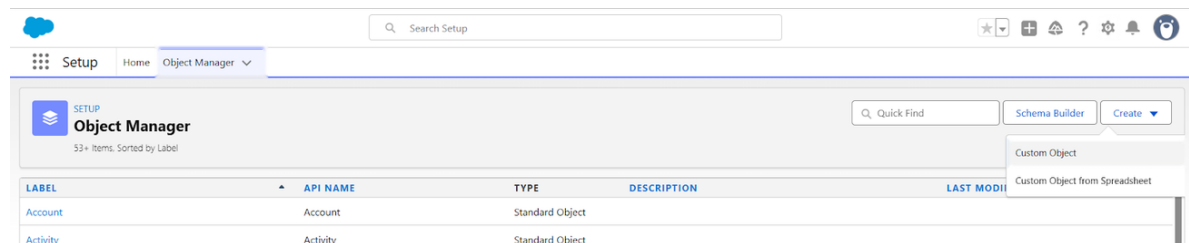
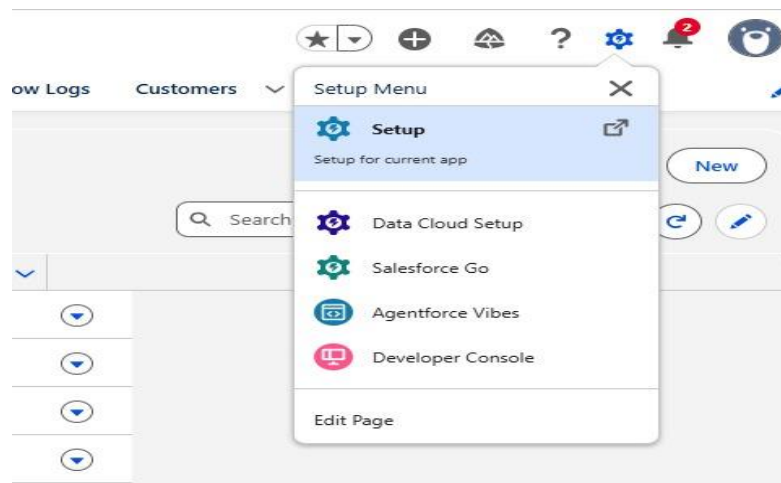
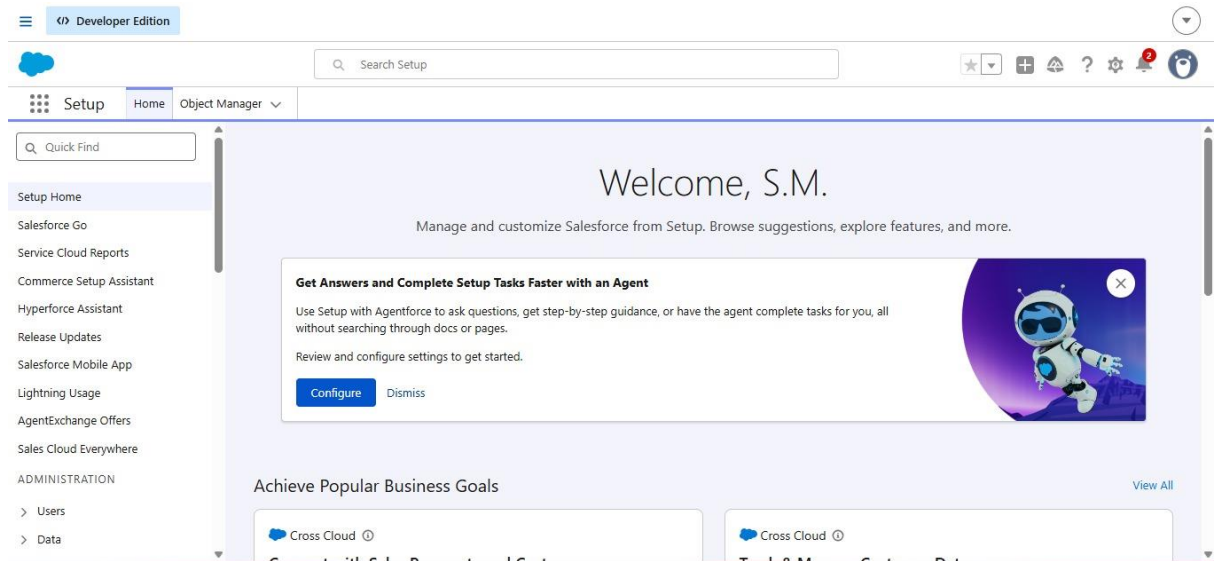


Figure 11: Object Manager – Create Custom Object

2. Enter Label: Customer, Plural Label: Customers and Record Name: Customer Name (Text).
3. Under Optional Features enable Allow Reports and Track Field History.

The screenshot displays the 'Optional Features' section of a Salesforce object setup page. It includes several sections with checkboxes and radio buttons:

- Optional Features:** Contains checkboxes for 'Allow Reports' (checked), 'Allow Activities' (checked), 'Track Field History' (checked), 'Allow in Chatter Groups' (unchecked), and 'Enable Licensing' (unchecked).
- Object Classification:** Includes a note about Enterprise vs. Light Application objects and checkboxes for 'Allow Sharing' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked).
- Deployment Status:** Features radio buttons for 'In Development' (unchecked) and 'Deployed' (checked).
- Search Status:** Includes a note about searchability and a checked checkbox for 'Allow Search'.
- Object Creation Options (Available only when custom object is first created):** Contains checkboxes for 'Add Notes and Attachments related list to default page layout' (checked) and 'Launch New Custom Tab Wizard after saving this custom object' (unchecked).

At the bottom, there are three buttons: 'Save', 'Save & New', and 'Cancel'.

Figure 12: Optional features and creation options for the object

4. In Search Status tick Allow Search.
5. Under Object Creation Options tick Add Notes and Attachments related list to default page layout.
6. Click Save.

Activity 2: Create the Shipment object

1. Go to Setup → Object Manager → Create → Custom Object.
2. Enter Label: Shipment and Plural Label: Shipments (Shipments).
3. Set Record Name: Tracking Number (Auto Number).
4. Use the display format SHP-{00000} with the starting number 0.
5. Enable Allow Reports and Track Field History.
6. Click Save.

Activity 3: Create the Delivery Agent object

1. Go to Setup → Object Manager → Create → Custom Object.
2. Enter Label: Delivery Agent and Plural Label: Delivery Agents (Delivery Agents).

3. Set Record Name: Agent Name (Text).
4. Enable Allow Reports and Track Field History.
5. Click Save.

Activity 4: Create the Branch object

1. Go to Setup → Object Manager → Create → Custom Object.
2. Enter Label: Branch and Plural Label: Branchs (Branches).
3. Set Record Name: Branch Name (Text).
4. Enable Allow Reports.
5. Click Save.

Activity 5: Create the Delivery Update object

1. Go to Setup → Object Manager → Create → Custom Object.
2. Enter Label: Delivery Update and Plural Label: Delivery Updates (Delivery Updates).
3. Set Record Name: Update Number (Auto Number).
4. Use the display format UPD-{0000} with the starting number 0.
5. Enable Allow Reports and Track Field History.
6. Click Save.

Activity 6: Create the Invoice object

1. Go to Setup → Object Manager → Create → Custom Object.
2. Enter Label: Invoice and Plural Label: Invoices (Invoices).
3. Set Record Name: Invoice Number (Auto Number).
4. Use the display format INV-{0000} with the starting number 0.
5. Enable Allow Reports and Track Field History.
6. Click Save.

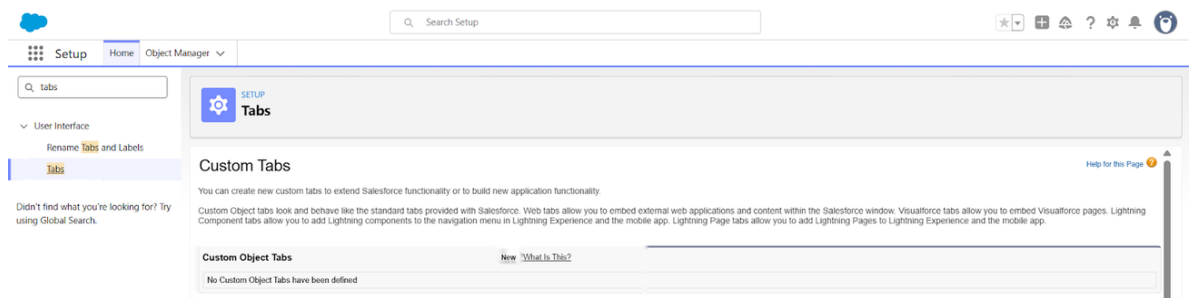
Milestone 3: Tabs

What is a tab?

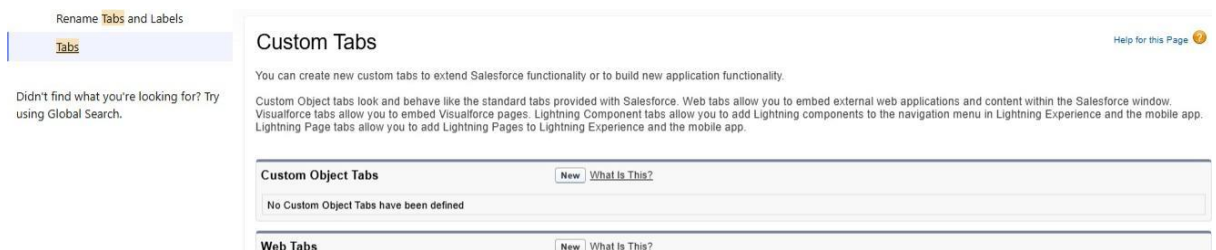
Tabs let users see the records of an object at a glance. They display the data of objects and other web content inside the application.

Activity 1: Create a tab for the Customer object

1. Click Home, type Tabs in Quick Find and select Tabs.

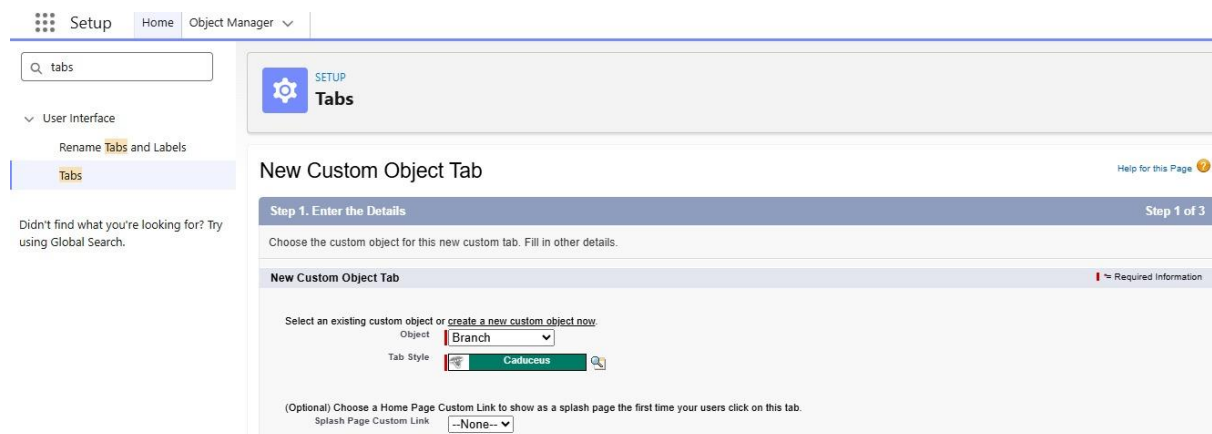


2. Under Custom Object Tabs click New.



3. For Object, select Customer

4. Choose any tab style.



5. Leave the other settings as they are.

6. Click Next, Next and Save.



Activity 2: Create a tab for the Shipment object

1. Click Home, type Tabs in Quick Find and select Tabs.
2. Under Custom Object Tabs click New.
3. For Object, select Shipment.
4. Choose any tab style.
5. Leave the other settings as they are.
6. Click Next, Next and Save.

Activity 3: Create a tab for the Delivery Agent object

1. Click Home, type Tabs in Quick Find and select Tabs.
2. Under Custom Object Tabs click New.
3. For Object, select Delivery Agent.
4. Choose any tab style.
5. Leave the other settings as they are.
6. Click Next, Next and Save.

Activity 4: Create a tab for the Branch object

1. Click Home, type Tabs in Quick Find and select Tabs.
2. Under Custom Object Tabs click New.
3. For Object, select Branch.
4. Choose any tab style.
5. Leave the other settings as they are.
6. Click Next, Next and Save.

Activity 5: Create a tab for the Delivery Update object

1. Click Home, type Tabs in Quick Find and select Tabs.
2. Under Custom Object Tabs click New.
3. For Object, select Delivery Update.
4. Choose any tab style.
5. Leave the other settings as they are.
6. Click Next, Next and Save.

Activity 6: Create a tab for the Invoice object

1. Click Home, type Tabs in Quick Find and select Tabs.
2. Under Custom Object Tabs click New.
3. For Object, select Invoice.
4. Choose any tab style.
5. Leave the other settings as they are.
6. Click Next, Next and Save.


SETUP
Tabs

Custom Tabs Help for this Page

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Custom Object Tabs		New	What Is This?
Action	Label	Tab Style	Description
Edit Del	Branches	 Caduceus	
Edit Del	Customers	 People	
Edit Del	Delivery Agents	 Box	
Edit Del	Delivery Updates	 Bell	
Edit Del	Invoices	 Alarm clock	

Milestone 4: Fields & Relationships

What are fields?

Fields are the columns of an object, just like columns in a relational database. Each field stores one kind of value for a record.

Activity 1: Fields of the Customer object

Field Name	API Name	Data Type	Required	Description
Customer Name	Name	Text	Yes	Full name of customer
Email	Email__c	Email	Yes	Customer email
Phone	Phone__c	Phone	Yes	Contact number
Address	Address__c	Text Area	Yes	Customer address
Customer Type	Customer_Type__c	Picklist	Yes	Regular / Business
City	City__c	Text	No	Customer city
State	State__c	Text	No	Customer state
Pincode	Pincode__c	Text	No	Postal code
Is Active	Is_Active__c	Checkbox	No	Active status

- **Email (Email):** Setup → Object Manager → Customer__c → Fields & Relationships → New. Select Email → Next. Field Label: Email. Tick Required. Click Next → Next → Save.
- **Phone (Phone):** Setup → Object Manager → Customer__c → Fields & Relationships → New. Select Phone → Next. Field Label: Phone. Tick Required. Click Next → Next → Save.
- **Address (Text Area):** Setup → Object Manager → Customer__c → Fields & Relationships → New. Select Text Area → Next. Field Label: Address. Tick Required. Click Next → Next → Save.
- **Customer Type (Picklist):** Setup → Object Manager → Customer__c → Fields & Relationships → New. Select Picklist → Next. Field Label: Customer Type. Enter the values Regular and Business. Tick Required. Click Next → Next → Save.
- **City (Text):** Setup → Object Manager → Customer__c → Fields & Relationships → New. Select Text → Next. Field Label: City. Click Next → Next → Save.
- **State (Text):** Setup → Object Manager → Customer__c → Fields & Relationships → New. Select Text → Next. Field Label: State. Click Next → Next → Save.

- **Pincode (Text):** Setup → Object Manager → Customer__c → Fields & Relationships → New. Select Text → Next. Field Label: Pincode. Click Next → Next → Save.
- **Is Active (Checkbox):** Setup → Object Manager → Customer__c → Fields & Relationships → New. Select Checkbox → Next. Field Label: Is Active. Click Next → Next → Save.

Activity 2: Fields of the Shipment object

Field Name	API Name	Data Type	Required	Description
Tracking Number	Name	Auto Number	Yes	Unique ID
Customer	Customer__c	Lookup (Customer)	Yes	Related customer
Source Address	Source_Address__c	Text Area	Yes	Pickup location
Destination Address	Destination_Address__c	Text Area	Yes	Delivery location
Weight	Weight__c	Number	Yes	Parcel weight
Price	Price__c	Currency	Yes	Shipping charge
Status	Status__c	Picklist	Yes	Shipment status
Expected Date	Expected_Date__c	Date	Yes	Delivery estimate
Assigned Agent	Assigned_Agent__c	Lookup (Delivery Agent)	No	Delivery agent
Shipment Type	Shipment_Type__c	Picklist	Yes	Domestic / International

- **Source Address (Text Area):** Setup → Object Manager → Shipment__c → Fields & Relationships → New. Select Text Area → Next. Field Label: Source Address. Tick Required. Click Next → Next → Save.
- **Destination Address (Text Area):** Setup → Object Manager → Shipment__c → Fields & Relationships → New. Select Text Area → Next. Field Label: Destination Address. Tick Required. Click Next → Next → Save.
- **Weight (Number):** Setup → Object Manager → Shipment__c → Fields & Relationships → New. Select Number → Next. Field Label: Weight. Tick Required. Click Next → Next → Save.
- **Price (Currency):** Setup → Object Manager → Shipment__c → Fields & Relationships → New. Select Currency → Next. Field Label: Price. Tick Required. Click Next → Next → Save.
- **Status (Picklist):** Setup → Object Manager → Shipment__c → Fields & Relationships → New. Select Picklist → Next. Field Label: Status. Enter the values Booked, In Transit, Out for Delivery, Delivered, Delayed and Cancelled. Tick Required. Click Next → Next → Save.

- **Expected Date (Date):** Setup → Object Manager → Shipment__c → Fields & Relationships → New. Select Date → Next. Field Label: Expected Date. Tick Required. Click Next → Next → Save.
- **Shipment Type (Picklist):** Setup → Object Manager → Shipment__c → Fields & Relationships → New. Select Picklist → Next. Field Label: Shipment Type. Enter the values Domestic and International. Tick Required. Click Next → Next → Save.

Activity 3: Fields of the Delivery Agent object

Field Name	API Name	Data Type			Required	Description
Agent Name	Name	Text			Yes	Name of the agent
Email	Email__c	Email			Yes	Agent email
Phone	Phone__c	Phone			Yes	Agent contact number
Branch	Branch__c	Lookup (Branch)			Yes	Home branch
Status	Status__c	Picklist			Yes	Availability
Joining Date	Joining_Date__c	Date	No	Date of joining		

- **Email (Email):** Setup → Object Manager → Delivery_Agent__c → Fields & Relationships → New. Select Email → Next. Field Label: Email. Tick Required. Click Next → Next → Save.

The screenshot shows the Salesforce Setup interface. The top navigation bar includes 'Developer Edition' and a search bar. The left sidebar shows 'Setup' with sub-items 'Home' and 'Object Manager'. The main content area is titled 'Object Manager' and shows a list of objects. The 'Customer' object is selected, showing its details: 'Customer__c' (Custom Object), '9/27/2026' (Last Modified), and 'Deployed' (checked). A search bar with 'customer' and buttons for 'Schema Builder' and 'Create' are also visible.

The screenshot shows the 'Fields & Relationships' section for the 'Customer' object. The left sidebar shows 'Details' and 'Fields & Relationships'. The main content area is titled 'Fields & Relationships' and shows a list of fields. The 'Email' field is selected, showing its details: 'Email__c' (Email), '9/27/2026' (Last Modified), and 'Deployed' (checked). A search bar with 'Quick Find' and buttons for 'New', 'Deleted Fields', 'Field Dependencies', and 'Set History Tracking' are also visible.

The screenshot shows the 'Email' field details for the 'Customer' object. The left sidebar shows 'Details' and 'Fields & Relationships'. The main content area is titled 'Email' and shows the field's details: 'Email__c' (Email), '9/27/2026' (Last Modified), and 'Deployed' (checked). The field's description is: 'Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.' A search bar with 'Quick Find' and buttons for 'New', 'Deleted Fields', 'Field Dependencies', and 'Set History Tracking' are also visible.

SETUP > OBJECT MANAGER
Customer

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters

Step 2. Enter the details Step 2 of 4

Field Label ⓘ

Field Name ⓘ

Description

Help Text ⓘ

Required ☒ Always require a value in this field in order to save a record
 Unique ☐ Do not allow duplicate values
 External ID ☐ Set this field as the unique record identifier from an external system
 Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity ⓘ

Setup Home Object Manager

SETUP > OBJECT MANAGER
Customer

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters

Customer
New Custom Field Help for this Page

Step 3. Establish field-level security Step 3 of 4

Field Label
 Data Type
 Field Name
 Description

Select the profiles to which you want to grant edit access to this field via field-level security. The field will be hidden from all profiles if you do not add it to field-level security.

Field-Level Security for Profile	Visible	Read-Only
Analytics Cloud Integration User	☒	☐
Analytics Cloud Security User	☒	☐
Anypoint Integration	☒	☐

Customer
New Custom Field Help for this Page

Step 4. Add to page layouts Step 4 of 4

- **Phone (Phone):** Setup → Object Manager → Delivery_Agent__c → Fields & Relationships → New. Select Phone → Next. Field Label: Phone. Tick Required. Click Next → Next → Save.
- **Status (Picklist):** Setup → Object Manager → Delivery_Agent__c → Fields & Relationships → New. Select Picklist → Next. Field Label: Status. Enter the values Active, Inactive and On Leave. Tick Required. Click Next → Next → Save.
- **Joining Date (Date):** Setup → Object Manager → Delivery_Agent__c → Fields & Relationships → New. Select Date → Next. Field Label: Joining Date. Click Next → Next → Save.

Activity 4: Fields of the Branch object

Field Name	API Name	Data Type	Required
Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes
Manager Name	Manager_Name__c	Text	No

For each field: Setup → Object Manager → Branch__c → Fields & Relationships → New, select the data type, enter the field label, set Required where shown and click Save. Repeat for all fields.

Activity 5: Fields of the Delivery Update object

Field Name	API Name	Data Type	Required
Update Number	Name	Auto Number	Yes
Shipment	Shipment__c	Master-Detail	Yes
Status	Status__c	Picklist	Yes
Update Date	Update_Date__c	Date/Time	Yes
Remarks	Remarks__c	Text	No

Status picklist values: In Transit, Out for Delivery, Delivered, Delayed. Tick Required and Save.

Activity 6: Fields of the Invoice object

Field Name	API Name	Data Type	Required
Invoice Number	Name	Auto Number	Yes
Shipment	Shipment__c	Lookup	Yes
Amount	Amount__c	Currency	Yes
Invoice Date	Invoice_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

Payment Status values: Paid, Unpaid, Pending. Tick Required and Save.

Relationships

Child Object	Field	Parent Object	Type
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery Agent	Lookup
Delivery Agent	Branch__c	Branch	Lookup
Delivery Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Relationship creation steps

Activity 1: Shipment → Customer

1. Setup → Shipment__c → Fields & Relationships → New.
2. Select Lookup Relationship.
3. Related Object: Customer__c.
4. Field Label: Customer.
5. Save.

Activity 2: Shipment → Delivery Agent

1. Setup → Shipment__c → New.
2. Select Lookup Relationship.
3. Related Object: Delivery_Agent__c.
4. Field Label: Assigned Agent.
5. Save.

Activity 3: Delivery Agent → Branch

1. Setup → Delivery_Agent__c → New.
2. Select Lookup Relationship.
3. Related Object: Branch__c.
4. Field Label: Branch.
5. Save.

Activity 4: Delivery Update → Shipment

1. Setup → Delivery_Update__c → New.
2. Select Master-Detail Relationship.
3. Related Object: Shipment__c.

4. Save.

Activity 5: Invoice → Shipment

1. Setup → Invoice__c → New.
2. Select Lookup Relationship.
3. Related Object: Shipment__c.
4. Save.

Milestone 5: Validation Rules

Validation rules are checked whenever a user tries to save a record. If the data does not meet the condition, an error message is shown and the record cannot be saved until the problem is corrected.

Activity 1: Validation rules on Customer__c

- **Email_Required:** Setup → Object Manager → Customer__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Email field. Save.

Formula: ISBLANK (Email__c)

Error message: Email is required.

- **Validate_Phone_Number:** Setup → Object Manager → Customer__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Phone field. Save.

Formula: LEN (Phone__c) <> 10

Error message: Phone number must be 10 digits.

- **Validate_Pincode:** Setup → Object Manager → Customer__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Pincode field. Save.

Formula: LEN (Pincode__c) <> 6

Error message: Pincode must be 6 digits.

Activity 2: Validation rules on Shipment__c

- **Validate_Weight:** Setup → Object Manager → Shipment__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Weight field. Save.

Formula: `Weight__c <= 0`

Error message: Weight must be greater than zero.

- **Validate_Price:** Setup → Object Manager → Shipment__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Price field. Save.

Formula: `Price__c <= 0`

Error message: Price must be greater than zero.

- **Validate_Expected_Date:** Setup → Object Manager → Shipment__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Expected Date field. Save.

Formula: `Expected_Date__c < TODAY()`

Error message: Expected delivery date cannot be in the past.

Activity 3: Validation rule on Delivery_Agent__c

- **Validate_Agent_Email:** Setup → Object Manager → Delivery_Agent__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Email field. Save.

Formula: `NOT(CONTAINS(Email__c, "@"))`

Error message: Enter a valid email address.

Activity 4: Validation rule on Branch__c

- **Branch_Code_Required:** Setup → Object Manager → Branch__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Branch Code field. Save.

Formula: `ISBLANK(Branch_Code__c)`

Error message: Branch Code is mandatory.

Activity 5: Validation rule on Delivery_Update__c

- **Remarks_Required_For_Delay:** Setup → Object Manager → Delivery_Update__c → Validation Rules → New. Tick Active, paste the formula, click Check Syntax, enter the error message and set the error location to the Remarks field. Save.

Formula: `AND(ISPICKVAL(Status__c, "Delayed"), ISBLANK(Remarks__c))`

Error message: Remarks are required when status is Delayed.

Milestone 6: Page Layouts

Page layouts control what a user sees on a record page and how the information is organised. They arrange fields, sections, buttons and related lists so that users can enter, update and view data easily.

Activity 1: Edit the page layout of Customer__c

1. Setup → Object Manager → Customer__c → Page Layouts.
2. Open the default Customer layout.
3. From the top palette drag Section onto the layout, enter the section name, choose a two-column layout with top-down tab order, and click OK.
4. Drag the fields listed below into the section.
5. Arrange the fields neatly and click Save.

Section 1 – Customer Information: Customer Name, Customer Type, Email, Phone, Is Active

Section 2 – Address Details: Address, City, State, Pincode

Activity 2: Edit the page layout of Shipment__c

1. Setup → Object Manager → Shipment__c → Page Layouts.
2. Open the default Shipment layout.
3. From the top palette drag Section onto the layout, enter the section name, choose a two-column layout with top-down tab order, and click OK.
4. Drag the fields listed below into the section.
5. Arrange the fields neatly and click Save.

Section 1 – Shipment Details: Tracking Number, Customer, Shipment Type, Weight, Price, Status

Section 2 – Address Information: Source Address, Destination Address

Section 3 – Delivery Information: Expected Date, Assigned Agent

Activity 3: Edit the page layout of Delivery_Agent__c

1. Setup → Object Manager → Delivery_Agent__c → Page Layouts.
2. Open the default Delivery Agent layout.
3. From the top palette drag Section onto the layout, enter the section name, choose a two-column layout with top-down tab order, and click OK.

4. Drag the fields listed below into the section.
5. Arrange the fields neatly and click Save.

Agent Details: Agent Name, Email, Phone, Branch, Status, Joining Date

Additional Information: Related lists: Shipments and Delivery Updates

Activity 4: Edit the page layout of Branch__c

1. Setup → Object Manager → Branch__c → Page Layouts.
2. Open the default Branch layout.
3. From the top palette drag Section onto the layout, enter the section name, choose a two-column layout with top-down tab order, and click OK.
4. Drag the fields listed below into the section.
5. Arrange the fields neatly and click Save.

Branch Information: Branch Name, Branch Code, City, State, Manager Name

Activity 5: Edit the page layout of Delivery_Update__c

1. Setup → Object Manager → Delivery_Update__c → Page Layouts.
2. Open the default Delivery Update layout.
3. From the top palette drag Section onto the layout, enter the section name, choose a two-column layout with top-down tab order, and click OK.
4. Drag the fields listed below into the section.
5. Arrange the fields neatly and click Save.

Update Information: Update Number, Shipment, Status, Update Date, Remarks

Activity 6: Edit the page layout of Invoice__c

1. Setup → Object Manager → Invoice__c → Page Layouts.
2. Open the default Invoice layout.
3. From the top palette drag Section onto the layout, enter the section name, choose a two-column layout with top-down tab order, and click OK.
4. Drag the fields listed below into the section.
5. Arrange the fields neatly and click Save.

Invoice Details: Invoice Number, Shipment, Amount, Invoice Date, Payment Status

Milestone 7: Flows

A flow is a powerful Salesforce tool for automating business processes, collecting and updating data, and guiding users through screens. Flows are built in a visual interface, so no coding is required.

Activity 1: Auto-update the shipment status when a delivery update is created

Scenario: Whenever a Delivery_Update__c record is created, the Status of the related Shipment__c record should automatically take the latest status from the update.

Part 1: Create the record-triggered flow

Step 1: Open Flow Builder

1. Click the gear icon → Setup.
2. Type Flows in Quick Find and open Flows.
3. Click New Flow.
4. Select Record-Triggered Flow and click Create.



Figure 14: Flows page in Setup

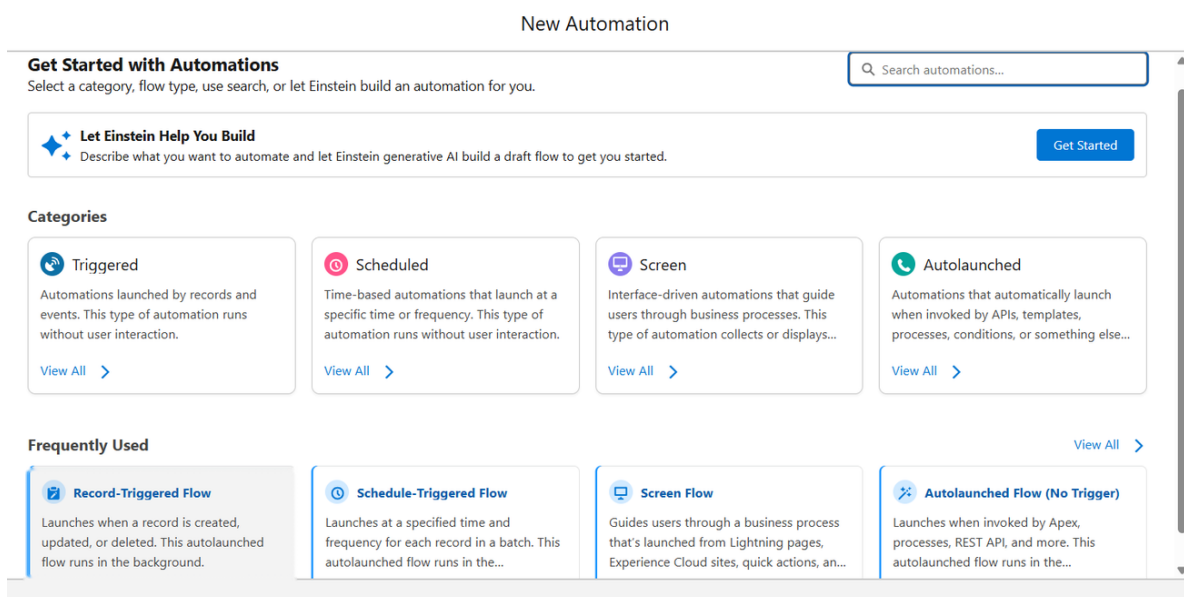


Figure 15: Selecting the Record-Triggered Flow type

Step 2: Configure the Start element

1. Object: Delivery_Update__c.
2. Trigger the flow when: A record is created.
3. Condition Requirements: None (always run the flow).
4. Optimise the flow for: Actions and Related Records.
5. Click Done.

Part 2: Update the related shipment

Step 3: Add the Update Records element

1. Click the + icon under Start and choose Update Records.
2. Label: Update_Shipment_Status.
3. Select Specify conditions to identify records.
4. Object: Shipment__c; Field: Id; Operator: Equals; Value: {\$Record.Shipment__c}.
5. Under Set Field Values choose Status__c and set the value to {\$Record.Status__c}.
6. Click Done.

Part 3: Save and activate

1. Click Save. Enter the flow label Auto_Update_Shipment_Status; the API name fills in automatically.
2. Click Activate and check that the status shows Active.

Auto-Update Shipment Status When Delivery Update is Created

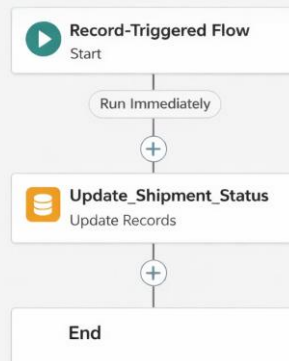


Figure 16: Completed flow layout

Milestone 8: The Lightning App

What is an app?

An app in Salesforce is a group of tabs that work together as one unit. It has a name, a logo and a set of tabs. The simplest app may have only two tabs.

Activity 1: Create the Courier Management System app

1. From Setup type App Manager in Quick Find and select App Manager.
2. Click New Lightning App and enter Courier Management System as the App Name.
3. Upload an image or icon for the app and click Next.
4. Keep the default App Options and click Next.
5. Keep the Utility Items as they are and click Next.
6. From Available Items move Customer, Shipment, Delivery Agent, Branch, Delivery Update, Invoice, Reports and Dashboards to Selected Items, then click Next.
7. From Available Profiles move System Administrator to Selected Profiles.
8. Click Save & Finish.

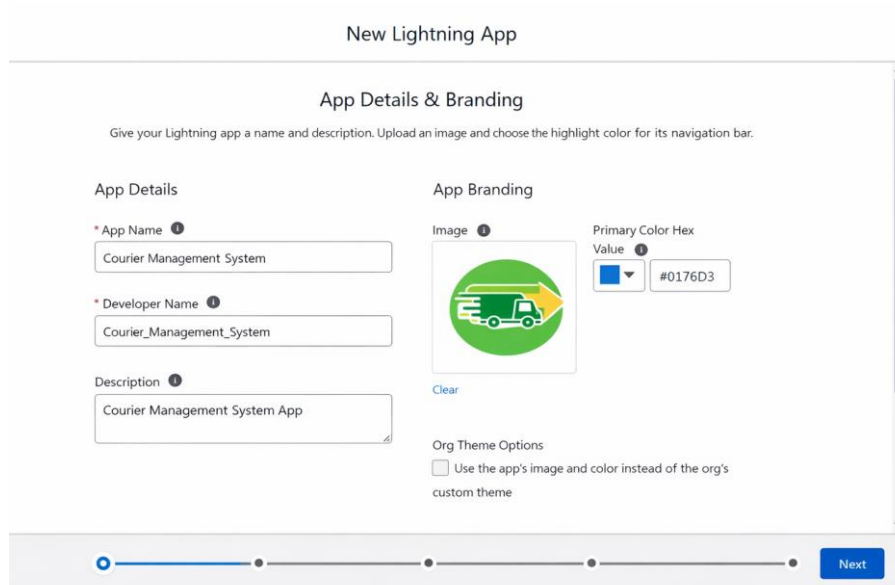


Figure 17: App details and branding

New Lightning App

User Profiles

Choose the user profiles that can access this app.

Available Profiles

Type to filter list...

Analytics Cloud Integration User

Analytics Cloud Security User

Anypoint Integration

Authenticated Website

Authenticated Website

B2B Reordering Portal Buyer Profile

▶

◀

Selected Profiles

System Administrator

Save & Finish

Figure 18: Choosing user profiles for the app

Milestone 9: Profiles

What is a profile?

A profile is a collection of settings and permissions that defines what a user can do in Salesforce. It controls object permissions, field permissions, user permissions, tab settings, app settings, Apex class and Visualforce page access, page layouts, record types, login hours and login IP ranges. A profile can be assigned to many users, but a user can have only one profile at a time.

Activity 1: Create the Courier Admin profile

1. From Setup enter Profiles in Quick Find and open Profiles.
2. Click New Profile.
3. Set Existing Profile to System Administrator.
4. Enter the Profile Name: Courier Admin.
5. Click Save.

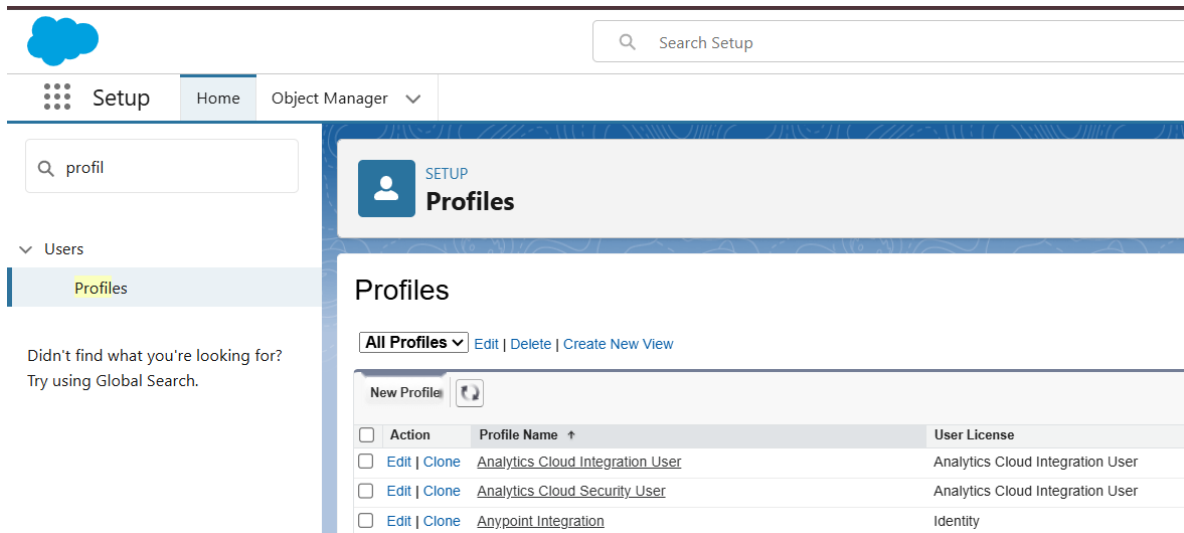


Figure 19: Profiles page in Setup

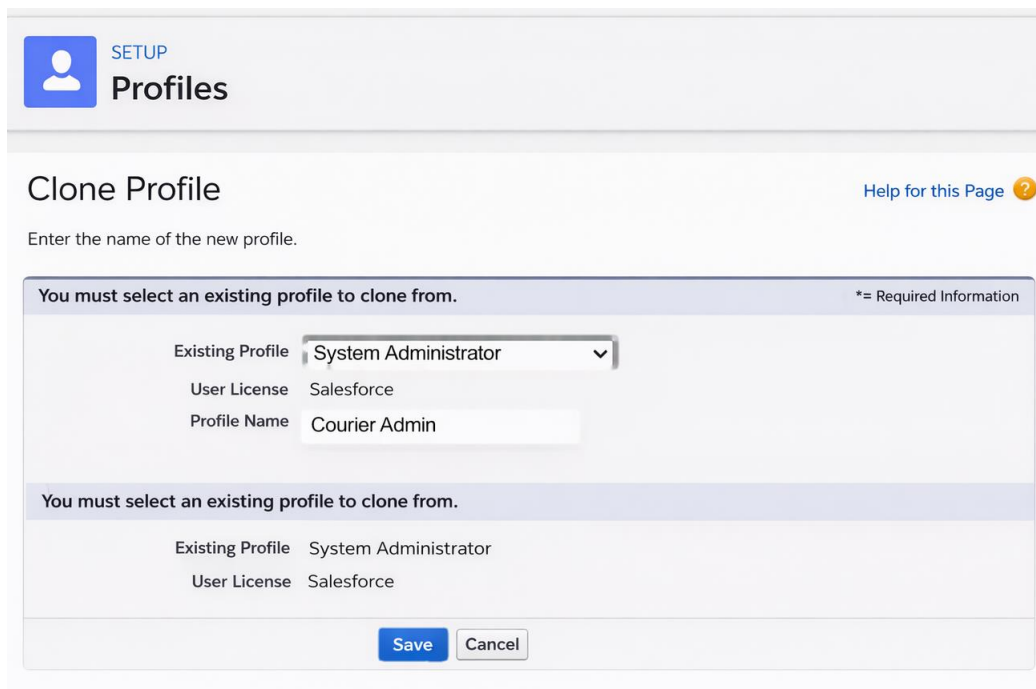


Figure 20: Cloning the System Administrator profile to create Courier Admin

Why use System Administrator as the base? It already has Modify All Data, Customize Application and full create-read-edit-delete access on all objects. By

cloning it, Courier Admin inherits full access automatically, and permissions can be restricted later if required.

Activity 2: Create the Branch Manager profile

1. Setup → Profiles → New Profile.
2. Existing Profile: Standard Platform User; Profile Name: Branch Manager; click Save.
3. Click Edit on the new profile.
4. Set the default app to Courier Management System.
5. Make these tabs visible: Customers, Shipments, Delivery Agents, Branches, Delivery Updates, Invoices, Reports and Dashboards.
6. Set the custom object permissions shown below.
7. Under Session Settings set the timeout to 2 hours of inactivity; under Password Policies set passwords to never expire with a minimum length of 8.
8. Click Save.

Object	Permissions
Customer__c	Create, Read, Edit
Shipment__c	Create, Read, Edit
Delivery_Agent__c	Read, Edit
Branch__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Invoice__c	Read

Custom Object Permissions

	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records 	Modify All Records 	View All Fields
Customer__c	☒	☒	☒	☒	☐	☐	☐
Shipment__c	☒	☒	☒	☒	☐	☐	☐
Delivery_Agent__c	☒	☒	☒	☐	☐	☐	☐
Branch__c	☒	☒	☒	☐	☐	☐	☐
Delivery_Update__c	☒	☒	☒	☒	☐	☐	☐
Invoice__c	☒	☐					

Figure 21: Custom object permissions configured for a profile

Activity 3: Create the Delivery Agent profile

1. Setup → Profiles → New Profile.
2. Existing Profile: Standard Platform User; Profile Name: Delivery Agent; click Save.
3. Set the default app to Courier Management System.
4. Make only the Shipments and Delivery Updates tabs visible.
5. Set the object permissions shown below.
6. Restrictions: no branch editing, no delete permission and limited reporting access.
7. Click Save.

Object	Permissions
Shipment__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Customer__c	Read
Invoice__c	Read

Activity 4: Create the Customer Support profile

1. Setup → Profiles → New Profile.
2. Existing Profile: Standard User; Profile Name: Customer Support.
3. Make the Customers, Shipments and Delivery Updates tabs visible.
4. Set the object permissions shown below.
5. Set the session timeout to 2 hours and the minimum password length to 8.
6. Click Save.

Object	Permissions
Customer__c	Read, Edit
Shipment__c	Read, Edit
Delivery_Update__c	Read
Invoice__c	Read

Final profiles created in the project

Profile Name	Purpose
Courier Admin	Full system control

Profile Name	Purpose
Branch Manager	Manage shipments and agents
Delivery Agent	Update delivery status
Customer Support	Handle customers and tracking

Milestone 10: Roles

Roles in Salesforce decide which users can access which data and functions, and they define the reporting hierarchy of the organisation. Users higher in the hierarchy have wider access to data and more control over the system.

Activity 1: Create the Courier Admin role

1. From Quick Find search for Roles and open Roles.
2. Click Set Up Roles.
3. Click Expand All and then Add Role under CEO.
4. Enter the label Courier Admin; the role name fills in automatically.
5. Click Save.

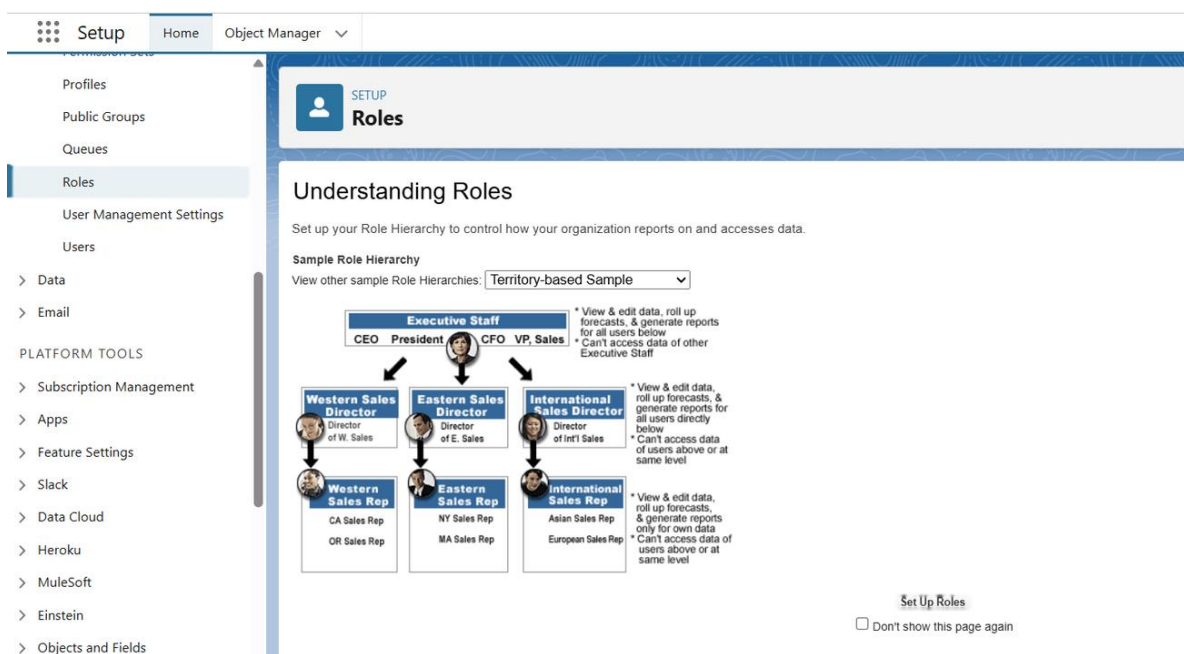


Figure 22: Roles page in Setup

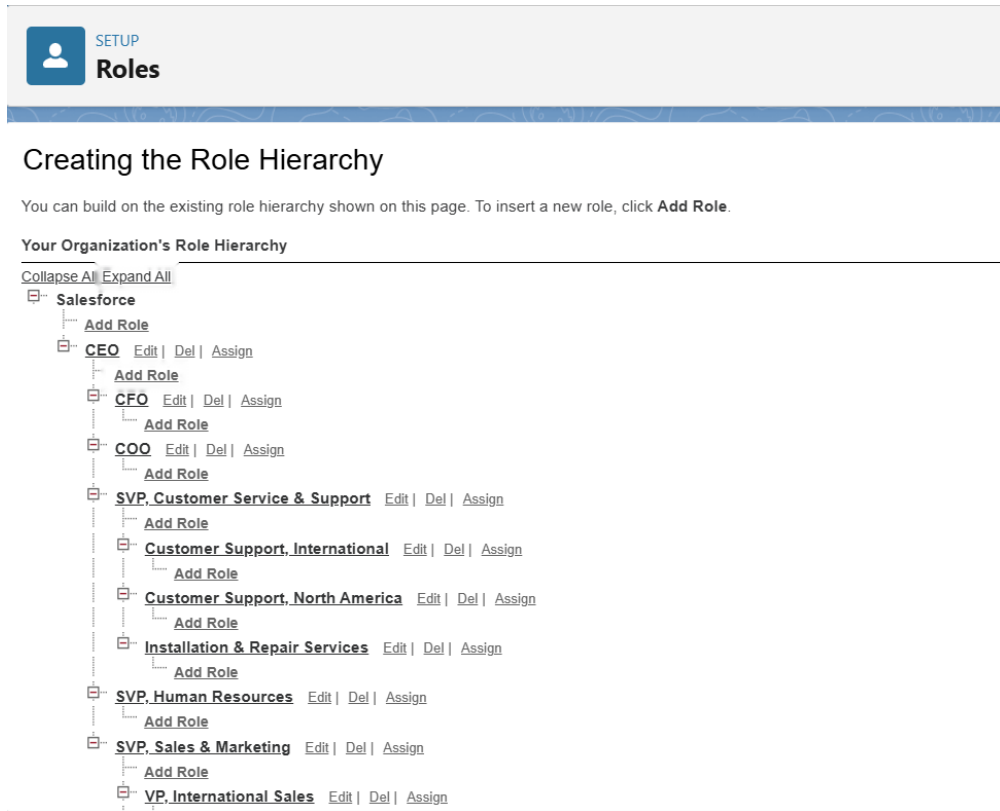


Figure 23: Role hierarchy creation screen

Activity 2: Create the Branch Manager role under Courier Admin

1. Below Courier Admin click Add Role.
2. Enter the label Branch Manager.
3. Click Save.

Activity 3: Create the Delivery Agent role under Branch Manager

1. Below Branch Manager click Add Role.
2. Enter the label Delivery Agent.
3. Click Save.

Activity 4: Create the Customer Support role under Branch Manager

1. Under Branch Manager click Add Role.
2. Enter the label Customer Support.
3. Click Save.

Purpose of the hierarchy

- **Courier Admin** – highest level, access to all data
- **Branch Manager** – manages branch-level operations
- **Delivery Agent** – access limited to assigned shipments

- **Customer Support** – limited to customer and shipment tracking

This hierarchy ensures proper data sharing and security across the Courier Management System.

Milestone 11: Users

What is a user?

A user is anyone who logs in to Salesforce, such as managers, staff and IT specialists who need access to company records. Every user has an account that identifies them, and the account settings decide which features and records they can use.

Note: A Developer org allows only a small number of full Salesforce licences. If none is available, deactivate a user who holds one, or change the licence type to another such as Salesforce Platform.

Activity 1: Create a user

1. From Setup enter Users in Quick Find and select Users.
2. Click New User.
3. Fill in the fields shown in the table below.
4. Click Save.

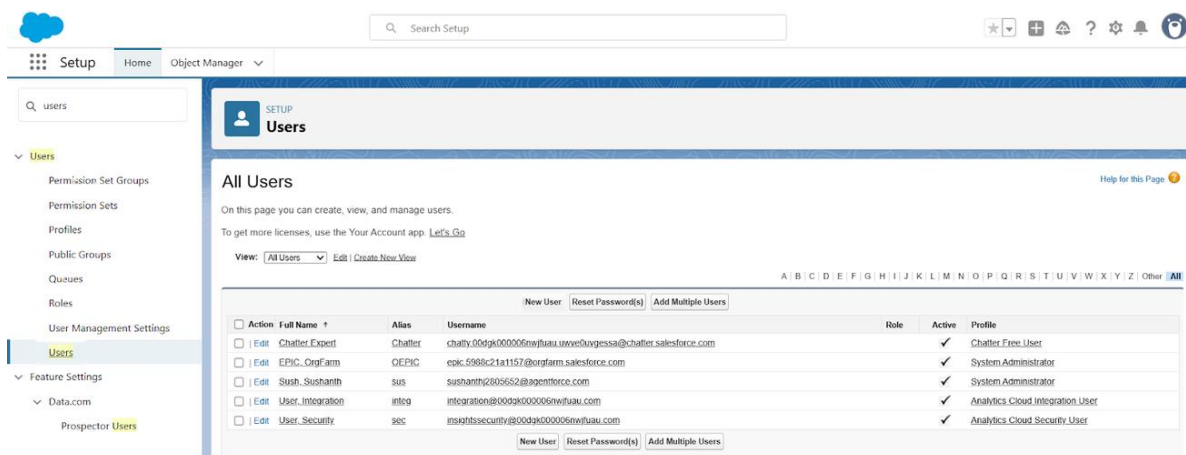


Figure 24: Users list in Setup

Field	Value
First Name	Enter first name (example: Ravi)
Last Name	Enter last name (example: Kumar)
Alias	Any short alias
Email	Your personal email address
Username	Unique, in the format text@text.text
Nickname	Any nickname
Role	Courier Admin

Field	Value
User License	Salesforce
Profile	Courier Admin

Activity 2: Create the other users

Repeat the same steps with the following details:

User Type	Role	User License	Profile
Branch Manager	Branch Manager	Salesforce	Branch Manager
Delivery Agent	Delivery Agent	Salesforce Platform	Delivery Agent
Customer Support	Customer Support	Salesforce Platform	Customer Support

Example – Branch Manager user: First name Arjun, last name Reddy, email arjun@gmail.com, username arjun.manager@cms.com, role Branch Manager, profile Branch Manager.

Important points

- Every user needs a unique username, a valid email, a role and a profile.
- Profiles control permissions.
- Roles control data visibility.

Milestone 12: Reports & Dashboards

Reports and dashboards help analyse business data and follow courier operations efficiently.

Activity 1: Create report types

1. Click the gear icon → Setup.
2. In Quick Find search Report Types.
3. Click New Custom Report Type.
4. Create the three report types in the table below; for each one set Category to Reports and Deployment Status to Deployed, then click Next → Save.

Primary Object	Label
Shipment__c	Shipment Tracking Report
Customer__c	Customer Summary Report
Delivery_Agent__c	Agent Performance Report

Activity 2: Create reports

Report 1: Shipment Status Summary

1. Open the App Launcher, search Reports and click New Report.
2. Choose Shipment Tracking Report and click Continue.
3. Add the columns Tracking Number, Customer, Status, Expected Date and Assigned Agent.
4. Filters: Show Me All Shipments; Date Filter This Month; optionally a Status filter.
5. Group rows by Status and add a summary on Price.
6. Click Save & Run. Name: Shipment Status Summary; Folder: CMS Reports.

Report 2: Agent Performance Report

1. New Report → choose Agent Performance Report → Continue.
2. Add the columns Agent Name, Branch, Number of Shipments and Status.
3. Group by Delivery Agent.
4. Save & Run. Name: Agent Performance Report; Folder: CMS Reports.

Report 3: Revenue by Branch Report

1. Create a report using the Shipment Tracking Report type.
2. Add the fields Tracking Number, Customer, Price and Invoice Status.
3. Group by Branch and Month.
4. Save as Revenue by Branch Report.

Activity 3: Create the dashboard

1. Open the App Launcher and search Dashboards.
2. Click New Dashboard.
3. Name: Courier Management Dashboard; Folder: CMS Reports; click Create.

Activity 4: Add dashboard components

1. Component 1 – Shipment Status Overview: click + Component, choose the Shipment Status Summary report, select Donut Chart and click Add.
2. Component 2 – Agent Performance: choose the Agent Performance Report, select Bar Chart and click Add.
3. Component 3 – Revenue by Branch: choose the Revenue by Branch Report, select Column Chart and click Add.

Activity 5: Save and share the dashboard

1. Click Save and then Done.
2. Share the dashboard with Courier Admin, Branch Managers and the Customer Support team.

Final layout



Figure 25: Final reports and Courier Management Dashboard

7. Functional & Performance Testing

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation and dashboard visualisation. All scenarios were validated to make sure the system performs correctly and reliably.

No.	Scenario	Expected Result	Status
1	Create a customer with a 9-digit phone number	Save blocked; "Phone number must be 10 digits."	Pass
2	Create a shipment with weight 0 or a past expected date	Save blocked; the matching error is shown	Pass
3	Add a Delayed delivery update without remarks	Save blocked; remarks required	Pass
4	Create a delivery update with status Delivered	Related shipment status changes to Delivered	Pass
5	Create an invoice for a delivered shipment	Invoice saved and linked to the shipment	Pass
6	Log in as Delivery Agent	Only Shipments and Delivery Updates tabs visible	Pass
7	Open the Courier Management Dashboard	Charts show shipment status, agent workload and branch revenue	Pass

8. Advantages & Disadvantages

Advantages

- All courier data is stored in one place
- Workflows are automated
- Reports give real-time insight

Disadvantages

- The solution depends on the Salesforce platform
- It needs some initial configuration effort

9. Conclusion

The Courier Management System successfully moves courier operations onto Salesforce. By combining a structured data model, validation, flow automation, role-based security and analytics, it improves efficiency and accuracy and gives customers and managers the visibility they need.

10. Future Scope

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimisation

11. Appendix

- Source code: Salesforce configuration
- Dataset: Salesforce objects
- GitHub / demo: Salesforce Developer org

Thank You